Evaluating Predictive Skeleton Learning for Video-Based Neurological Gait Analysis

Abstract

Multiple sclerosis (MS) and Parkinson's disease (PD) are neurological conditions associated with gait abnormalities, such as variations in stride, balance, and coordination. We investigate whether a self-supervised skeleton representation can distinguish gait videos labeled as Normal, MS, or PD. We compiled 88 gait sequences from 41 distinct YouTube sources and extracted 33 keypoints per frame using Google's MediaPipe BlazePose to train a Skeleton Joint Embedding Predictive Architecture (S-JEPA) with stochastic regional masking guided by 12 clinically selected landmarks. The frozen encoder was evaluated using five source-grouped outer folds and compared against mean pose, keypoint-visibility statistics, an engineered-feature Random Forest, and a majority-class baseline. S-JEPA achieved the highest average F1 score across the three target classes (0.468 vs. 0.452 for mean pose and 0.411 for RF), although bootstrap analysis left both advantages uncertain. Mean pose performed better under clip weighting and in four of the five folds. On this small-scale dataset, the findings do not provide sufficient evidence that temporal representation learning offers an advantage over simpler pose-based representations.

1. Introduction

Gait depends on the coordinated interaction between the motor, sensory, cerebellar, spinal cord, and brainstem systems. Neurological disorders can disrupt this coordination, resulting in altered gait parameters. Patients with multiple sclerosis (MS) demonstrate a variety of gait abnormalities rather than a single typical pattern due to varying location and severity of lesions [1–4]. Observed changes include shorter step length, decreased knee and ankle range of motion, greater gait variability, longer double-support time, and impaired bilateral coordination. Some of these impairments are associated with disability levels or become more pronounced with prolonged walking [5–10]. Parkinson's disease (PD) causes a number of overlapping gait abnormalities, including decreased stride length, increased variability, freezing, and gait asymmetry [11–14].

Quantitative gait assessment typically relies on specialized systems with measuring devices, wearable sensors, or motion capture laboratories. Although these methods allow for accurate biomechanical measurements, they are often impractical for evaluation outside of specialized clinical settings. Smartphone and video-based methods are other existing methods currently being investigated for gait monitoring and neurological disease assessment [15–21]. Current body pose estimation systems, such as OpenPose, HRNet, and BlazePose, can extract joint trajectories directly from ordinary videos [26–28]. However, these estimators are sensitive to camera angles, occlusions, frame rates, detector reliability, and preprocessing methods [22–25].

Therefore, the model may learn recording-specific or static information rather than meaningful gait dynamics.

Recent skeleton-based models have used graph neural networks, transformers, and self-supervised learning to represent human motion [29–34]. Studies show that even without class labels, useful representations can be learned by predicting hidden features from observed states [35–42]. These methods are promising for identifying temporal relationships between joints. However, the effectiveness of this method on small clinical datasets remains uncertain; such datasets may contain few independent video recordings, and repeated segments may lead to overfitting. Under such conditions, source-grouped evaluation is necessary to prevent related recordings from appearing in both training and test sets; moreover, the model selection process should be independent of the final evaluation [43,44].

The aim of this study was to investigate whether a self-supervised temporal representation of skeletal points can provide a consistent classification advantage over simpler pose-based summaries on a small, heterogeneous video dataset. We evaluate an encoder with a compact S-JEPA architecture using data from 88 usable video clips from 41 source recordings labeled Normal, MS, or PD. The encoder is pre-trained without class labels, frozen, and evaluated using source-grouped cross-validation. We compare the representations generated by this encoder with the following benchmarks: average pose, keypoint visibility statistics, random forest model based on engineered features, and majority class baseline.

The results show that the trained representation does not have a clear overall advantage over the average pose. Although their overall performance is similar, differences are apparent when clips or sources are given equal weights. Therefore, these results do not prove that learning using temporal representations improves classification performance on the current dataset. Furthermore, the interpretation of the results is limited by factors such as varying data collection conditions, unverified participant information in the original recordings, the shortcomings of the random forest feature extraction tool, and the mismatch between mirroring and masking operations during S-JEPA training. This study therefore provides an empirical evaluation of S-JEPA against simpler baselines and identifies the controlled comparisons needed to determine whether its learned representations capture predictive gait information beyond static pose and recording-related characteristics.

2. Related Work

2.1 Video-Based Gait Analysis

Pose estimators like BlazePose [28] allow for the extraction of joint trajectories from ordinary video, without markers or specialized equipment. Stenum et al. have validated such 2D pose-based gait measures against marker-based motion capture, finding good agreement across

most spatiotemporal parameters and expanding the validation across multiple camera perspectives and clinical populations [22], [23]. These works generally find pose estimation to be most reliable for sagittal-plane, lower-limb kinematics and simple spatiotemporal measures, and less reliable under conditions of occlusion, poor camera angle, or low frame rate that affect keypoint tracking.

The general validity of this approach has since been applied directly to neurological gait assessment. Creagh et al. have used smartphone-based data to track MS progression over time [15], while Eguchi et al. have found that pose-based features can help distinguish PD from spinocerebellar degeneration, a condition with overlapping symptoms [18] -- a differential-diagnosis setting closely related to our own three-class classification problem. This body of work demonstrates the feasibility and clinical value of video-based gait assessment, and also raises a concern relevant to our approach: since pose estimators are sensitive to video recording conditions as well as the subject's gait, a model trained on heterogeneous video might learn to predict properties of the recording process rather than the subject's gait. This concern is the motivation for the source-grouped evaluation adopted in Section 5.2.

2.2 Skeleton-Based Representation Learning

Skeleton-based methods have demonstrated that joint-position data alone is sufficient to learn meaningful motion representations, without the need for raw video. Approaches such as ST-GCN [29] model the skeleton as a spatiotemporal graph, capturing spatial information between joints and temporal changes in motion, with later methods focusing on refining how information is shared between joints [31]. Recently, transformer-based methods such as SkeletonMAE have adapted masked-prediction pretraining (Section 2.3) to skeleton sequences directly [34].

Within gait, skeleton-based graph architectures have focused on gait recognition (identifying individuals by their walk) rather than clinical classification. Tian et al. combined spatiotemporal attention with graph convolution for abnormal gait recognition [49], the closest related work to ours, though framed as abnormality detection rather than diagnosis-specific classification. This literature demonstrates that skeleton-only representations retain sufficient information to classify gait patterns at the individual and abnormality level, motivating our use of a similar approach for MS and PD.

The original S-JEPA formulation selects a set of joints to mask according to highest motion magnitude: the joints that move the most within a clip are hidden, and the model is trained to predict them from the remaining, lower-motion joints [41], [42]. More recent work has turned towards masking based on anatomical structure, grouping joints by body region (e.g., arm, leg, torso) rather than selecting purely by motion [54], and anatomically-guided architectures have been applied to skeleton-based recognition of neurological conditions [55], though not tailored to the specific characteristics of individual conditions. These works provide context for the

landmark-selection strategy introduced in Section 4.1 and evaluated in this study. We use clinical literature to motivate a fixed set of masked joints, but do not directly compare this choice against motion-magnitude or alternative anatomical masks.

2.3 Predictive and Self-Supervised Representation Learning

A broader class of work has demonstrated that useful representations can be learned without labels by predicting hidden content from visible context. Masked autoencoders (MAE) reconstruct masked image patches from the visible region [36]. Wei et al. demonstrated that, rather than reconstructing raw pixel values, masked content prediction in feature space yields representations that better preserve semantic structure [38].

Most directly relevant to the present work are joint-embedding predictive architectures (JEPAs), which combine masked feature prediction with an additional target encoder that undergoes a separate momentum update. I-JEPA applies this setup to images [41], while following work extends the feature prediction to video [42]. Abdelfattah and Alahi introduced S-JEPA for skeleton sequences. Rather than reconstructing the coordinates of hidden joints, S-JEPA predicts learned feature representations of those joints from the visible parts of the sequence [57]. Building on this existing framework, our study evaluates S-JEPA for neurological gait classification using video-derived skeleton sequences.

Given our relatively small dataset compared with typical self-supervised pretraining corpora, we also consider methodological work on cross-validation with limited sample sizes. Varoquaux showed that cross-validation can yield highly variable error estimates with small samples [43], while Varma and Simon demonstrated that using cross-validation for both model selection and evaluation can introduce optimistic bias [44]. These findings motivate our use of source-grouped cross-validation with independent model selection in Section 5.2 and our cautious interpretation of the results in Section 7.

3. Dataset and Preprocessing

We compiled 91 MP4 gait clips spanning Normal, MS, and PD. Each filename includes an 11-character YouTube video ID. A clip is kept if at least 30% of its sampled frames contain every x-coordinate and at least eight frames survive cleaning. Each frame is centered on the hip midpoint and divided by the shoulder-to-hip-midpoint distance. Each surviving frame was processed with MediaPipe BlazePose [28], which outputs 33 body landmarks per frame as normalized (x, y) image coordinates together with a visibility score for each landmark. S-JEPA tokenizes all 33 landmarks, using their x-coordinates, y-coordinates, and visibility values. Its training masks are sampled over anatomically connected regions, with region-selection weights biased toward regions containing 12 clinically selected joints; this does not restrict the model's

input to those 12 joints. The mean-pose and visibility baselines also use all 33 landmarks, whereas the Random Forest baseline uses engineered hip, knee, and ankle angle features. Three clips failed pose extraction, leaving 88 clips from 41 sources. The metadata does not verify participant identity or clinical status, so we treat the labels as collection labels and the source ID as the grouping unit. Figure 1 summarizes the processing and evaluation pipeline.

Table 1. Dataset composition by collection label.

Collection label	Raw clips	Usable clips	Sources	Padded clips	32-frame windows
Normal	26	24	16	5	59
MS	30	29	13	0	171
PD	35	35	12	10	128
Total	91	88	41	15	358

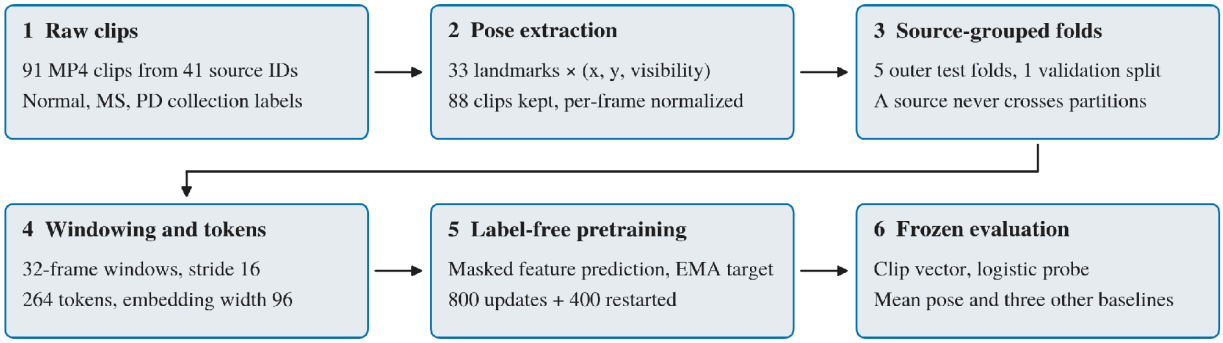


Figure 1. Pipeline from raw clips to held-out evaluation. Source partitions are fixed before windows are generated, so every clip, window, and augmented copy of a source stays in one partition.

Frames are subsampled with the integer stride $round(f_{src}/15)$. Nominal frame rates range from 23.76 to 59.94 fps, and clips span nine resolutions. After partitioning (Section 5), sequences are cut into 32-frame windows with stride 16 to provide a consistent input length for the S-JEPA encoder. The 15 clips shorter than 32 frames (10 PD, 5 Normal, no MS) are padded by repeating the last frame to reach the minimum window length.

4. Methodology

4.1 Stochastic Regional Masking and Sampling Bias

Rewrite: The masked-prediction objective requires choosing which tokens are hidden from the context encoder and predicted from the visible remainder. The original S-JEPA method uses motion-aware masking, which preferentially masks joints with greater motion [57]. Anatomically structured masking has also been explored in other skeleton-based self-supervised methods [54]. In this study, we use stochastic regional masking with a clinically motivated sampling bias for the gait-classification task.

A new mask is sampled for each training window. Connected body regions (head, mouth, each arm, trunk, and each leg) are hidden over contiguous spans of the eight time blocks until approximately 60% of the 264 tokens are masked. To reflect clinical findings of altered gait patterns in MS and PD [1, 8, 10, 11],we multiplied the sampling weights by 1.5 for regions containing any of 12 clinically selected landmarks: the bilateral shoulders, hips, knees, ankles, heels, and foot indices. In a bank of 5,000 sampled masks, the observed masking frequencies were 77–78% for shoulders and hips, 62–63% for other leg joints, 64% for elbows and wrists, and 58% for facial landmarks. These frequencies characterize the sampled masks rather than necessarily reflecting the frequencies encountered during training. No ablation compared this sampler with uniform, motion-based, or alternative anatomical masking, so its effect on classification performance remains unknown.

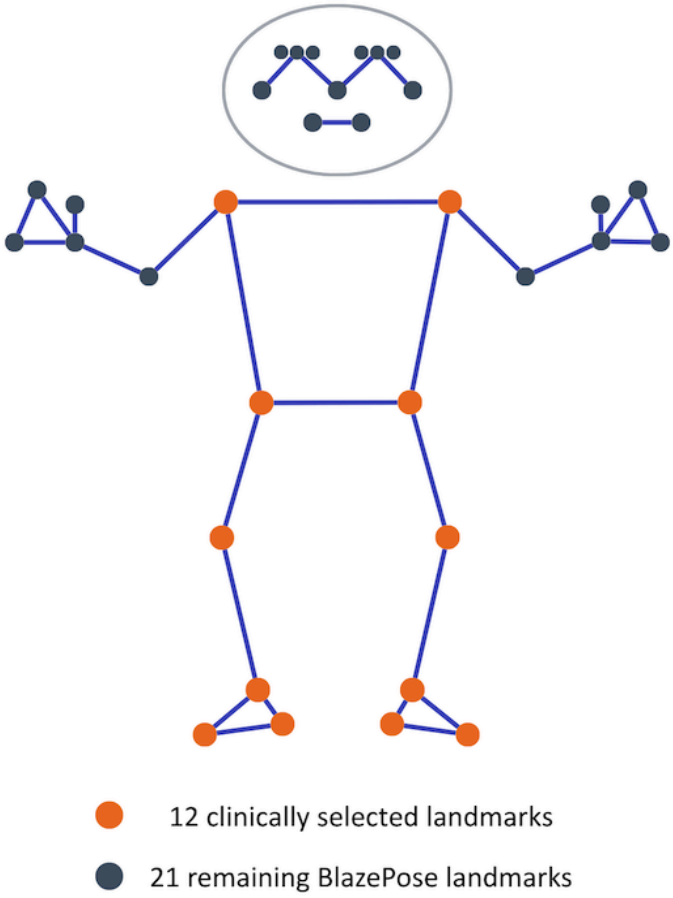


Figure 2. The 12 highlighted BlazePose landmarks (bilateral shoulders, hips, knees, ankles, heels, and foot indices) were selected based on clinical literature describing gait abnormalities associated with MS and PD. Regions containing these landmarks receive a 1.5× sampling weight during stochastic regional mask sampling. The remaining 21 landmarks are shown in gray. All 33 landmarks are used by S-JEPA; the highlighted landmarks do not constitute a fixed mask. Landmark definitions and skeletal connections are based on MediaPipe BlazePose [28]. No ablation was conducted to isolate the contribution of this sampling bias to downstream classification performance.

4.2 S-JEPA Architecture

Notebook 01 uses a MediaPipe estimator from the BlazePose family on full frames, keeping the first detected pose. Each frame has 33 landmarks with three channels: pixel x, pixel y, and visibility, the estimator's confidence that a landmark is visible. Depth was discarded. The sampling is approximately 15Hz rather than precisely aligned across recordings.

A clip must have finite x coordinates for all 33 landmarks in at least 30% of sampled frames and at least eight frames after cleaning. Empty ends are trimmed. Missing coordinates and visibility values are linearly interpolated within a clip, with no maximum gap length. The rejection log lists the three excluded clips, but not which rejection criterion each failed. Interpolation changes the observations and may smooth real irregularities.

Let p_t be the hip midpoint and s_t the shoulder midpoint. We normalize joint position x_{tj} by

$$\tilde{x}_{tj} = \frac{x_{tj} - p_t}{max(\|s_t - p_t\|_2, 10^{-3})}.$$

Visibility stays unchanged. A common positive scale preserves defined joint angles and length ratios in the estimated 2D pose within each frame. It removes absolute pelvis motion and apparent torso size. Since scale can change between frames, the amount of movement measured between frames also changes. The pipeline used therefore preserves landmark order and specific geometric relationships rather than the exact physical gait signal or every original movement exactly.

The framework deploys a Skeleton Joint Embedding Predictive Architecture (S-JEPA) tailored for skeletal time-series data. The model consists of two principal neural networks: a context encoder and a target encoder. Both of these use a transformer in order to convert the skeletal joint coordinates into a common latent space. The predictor network, which is connected to the context encoder directly in the latent space and thus avoids the requirement for pixel-level construction, relying entirely on the structural kinematic dynamics.

4.3 Training Strategy

Notebook 03 trains on all three conditions in the training partition without using their labels. We draw windows with replacement so every training source has equal total sampling probability. Within a source, its windows are equally likely. This prevents sources with many windows from dominating training, while changing the raw frequency distribution. The batch size is 32, and the initial stage has 800 updates.

The student receives a transformed view of each window: rotation within $\pm 15^{\circ}$, scale in $[0.9, 1.1]$, translation within ± 0.1 normalized units on each axis, and horizontal reflection with probability 0.5. One transform applies across the window, preserving sampled time order. Reflection swaps left/right landmark slots and their visibility. It retains angle magnitudes but exchanges sides. Because the teacher stays untransformed, correspondence between physical sides needs further testing; we do not claim that this augmentation preserves all useful asymmetry information.

For predictor vector u, teacher vector z, and running center c, training compares distributions over the 96 feature dimensions:

$$p = softmax(u/0.10), \qquad q = softmax((z - c)/0.06).$$

Cross-entropy $-\sum_d q_d log p_d$ is averaged over each example's hidden tokens, then over examples. These are feature distributions, not probabilities of the three conditions. Subtracting the center and sharpening the teacher distribution aim to prevent all inputs from receiving the same features. The lower teacher temperature concentrates probability on larger feature values. Once per batch, the center keeps 90% of its previous value and adds 10% of the masked target mean. The teacher receives no loss gradient; instead, an exponential moving average slowly moves its weights toward the student's, with momentum rising from 0.996 toward 1.

AdamW uses peak learning rate 10^{-3}, weight decay 0.05, coefficients $(0.9, 0.95)$, and gradient-norm clipping at 1. The learning rate rises for the first 10% of updates, then follows cosine decay. Model initialization and training use seed 42. Notebook 04 keeps the learned weights for another 400 updates on the same sources, but restarts the optimizer, running center, random-number generators, and schedules. This differs from uninterrupted 1,200-update training. Both stages use the same label-free feature-prediction loss.

Each stage gets its own classifier on frozen features. Validation selects the stage with higher source-weighted macro-F1, keeping the initial stage on a tie. Validation sources do not fit the encoder, scaler, or classifier, and there is no refit on combined training and validation data. We extract outer-test features after this choice is fixed.

4.4 Representation Readout

To gauge the utility of the learned latent representations for use in clinical settings, a frozen feature extraction approach was used. The weights of the context encoder are frozen. A linear classifier or lightweight multilayer perceptron (MLP) is trained on top of these frozen embeddings to classify subjects into healthy controls vs multiple sclerosis based strictly on kinematic movement patterns.

5. Experimental Setup

5.1 Baseline Methods

We compare the frozen S-JEPA encoder against four baselines. Random Forest (RF) uses hip, knee, and ankle joint angles computed from the cached landmarks: per-side means, left/right differences, and per-side ranges. Although the underlying feature container was designed for up to 82 engineered gait measures, only 15 columns vary across clips in this dataset; the rest are constant and removed before fitting. A known implementation defect causes left-ankle-range to be filled from the right ankle's statistics, so both ankle-range columns are identical across all 88 clips; this is not corrected in the results reported here.

Mean pose averages each landmark's root-centered, torso-scaled (x, y) coordinates over the full clip, discarding frame order. Keypoint-visibility statistics instead record each landmark's mean and standard deviation of detector visibility over the clip, using neither coordinates nor frame

order. Both use all 33 BlazePose landmarks, not the 16-landmark subset used for S-JEPA masking (Section 4.1), and both are classified with the same logistic-regression recipe used for the frozen S-JEPA features, fit independently on the training partition only.

The majority class predicts the most frequent label among the training clips in each fold, using no skeleton features.

5.2 Source-Grouped Evaluation

The five-fold partitioning of the 41 sources (seed 42) describes the partitioning of the dataset. For the k-th outer fold, the remaining sources are divided using a four-fold method. Only the first split, using seed 43+k, is used. Thus, the system has five outer folds, and each is associated with one inner validation holdout. Since this operation proceeds from the source to the clip and window, each clip and window receives its source partition (Table 2). Across the five outer folds, every usable clip appears in the test set exactly once.

Table 2. Source-grouped partitions by outer fold.

Fold	Training sources / clips	Validation sources / clips	Test sources / clips
0	24 / 51	8 / 19	9 / 18
1	24 / 58	9 / 17	8 / 13
2	24 / 54	9 / 17	8 / 17
3	24 / 53	9 / 21	8 / 14
4	24 / 45	9 / 17	8 / 26

Each row covers all 41 sources and 88 usable clips. Every clip is tested exactly once, and all three collection labels occur in every partition.

5.3 Evaluation Metrics

For each class k (Normal, MS, or PD), F1 is $2TP_k/(2TP_k + FP_k + FN_k)$, and macro-F1 takes the average of the three class values. The primary score is pooled source-weighted macro-F1. Clip j, whose source contributes n_j usable clips, receives weight $w_j = 1/n_j$. Pooled confusion counts and class F1 scores are

$$C_{kl} = \sum_{j:\, y_j=k,\, \hat{y}_j=l} w_j, \qquad F1_k = \frac{2C_{kk}}{\sum_l C_{kl} + \sum_l C_{lk}}, \tag{X}$$

where y_j and $\hat{y}_j$ are the label and prediction of clip j, and macro-F1 takes the average of $F1_k$ over the three classes. We also report clip weighting ($w_j = 1$) and the mean and population SD of the five per-fold scores.

6. Results

6.1 Training Diagnostics

We report the full-v1 cached run `eccab1697e1a8354`, with registry `9496e61b050f` and the laptop settings above. The accompanying evidence files record its hashes. Figure 3 shows saved log samples and stage endpoints. In fold 0, initial loss falls from 6.301 to 1.097, and effective rank ends at 12.13. Initial endpoint rank ranges from 10.91 to 12.13 across folds, while continuation endpoints range from 9.53 to 11.70. Effective rank estimates how many feature directions carry variation. We normalize the singular values of a centered feature batch, compute their entropy, and exponentiate it [4]. These diagnostics use centered batches of 32, giving a rank ceiling of 31 despite having 96 features. They show variation beyond a constant representation in those batches, without measuring condition information.

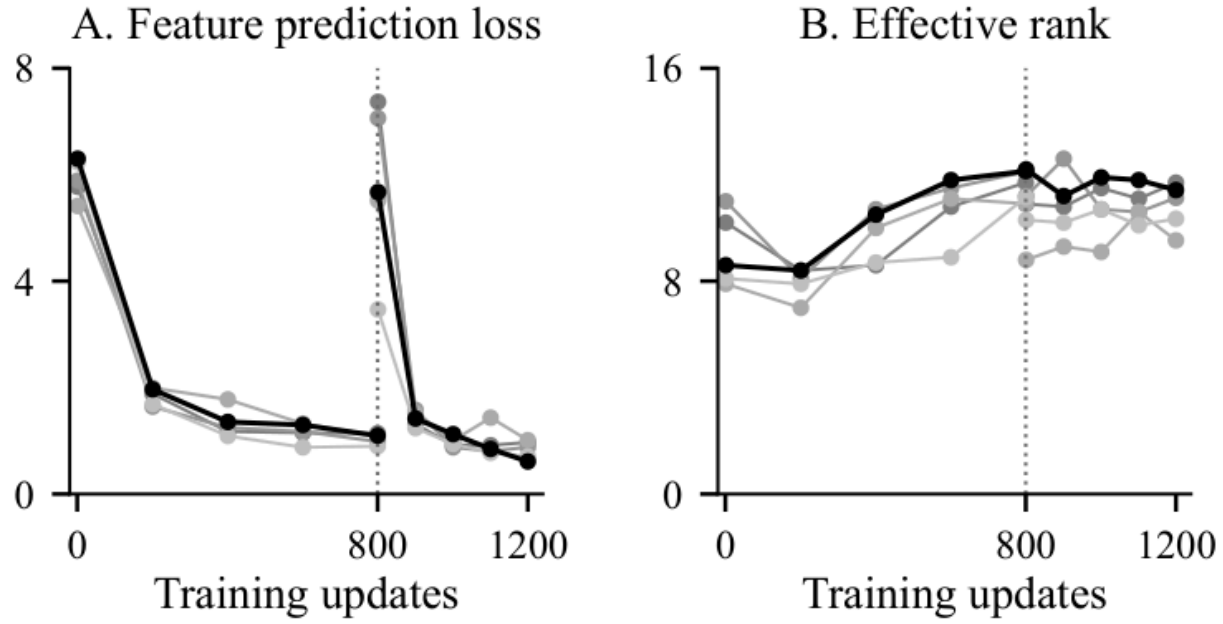


Figure 3. Training diagnostics. Black denotes fold 0; gray denotes the other folds. Points are sparse log samples and endpoints, joined for readability rather than full learning curves. The dotted line marks the restart at 800 updates. Resetting the optimizer, center, and schedules can produce a loss jump. Endpoint loss comes from one sampled training minibatch. Rank describes batch feature diversity, not accuracy.

Continuation lowers the final minibatch loss in every fold, but validation favors it only in folds 3 and 4. Fold 0 validation macro-F1 drops from 0.294 to 0.276; fold 4 improves from 0.375 to 0.560. The initial stage is selected in folds 0–2. Lower final training loss did not consistently correspond to higher validation macro-F1.

Notebook 05 provides a separate training-only check from its standalone fold-0 run, using 51 clips from 24 sources. After training-set standardization, silhouette scores are -0.0044 for initial embeddings, -0.0144 after continuation, and -0.0042 for visibility. Silhouette

compares closeness within a label group with closeness to other groups. Values near zero show little overall separation by that measure. These scores use the full features, rather than the two-dimensional visualization, and do not estimate test accuracy.

6.2 Classification Performance

All 88 clips have one verified outer-test prediction. S-JEPA has the highest pooled source-weighted macro-F1 (Table 2), exceeding RF by 0.057 and mean pose by 0.016. With equal clip weighting, mean pose slightly exceeds S-JEPA (0.410 versus 0.406). One MS source supplies 13 clips, so the choice of weights changes how strongly it affects the score. S-JEPA, RF, and mean pose each classify 37 of 88 clips correctly; their F1 differences reflect which sources and labels receive correct predictions.

Table 3. Held-out macro-F1. Pooled scores combine all folds before scoring. Fold mean ± SD uses source weighting. SD describes fold spread, not a confidence interval. RF retains the audited ankle-feature defect.

System	Pooled source	Pooled clip	Fold mean ± SD
S-JEPA	0.468	0.406	0.452±0.189
RF	0.411	0.397	0.395±0.098
Mean pose	0.452	0.410	0.427±0.132
Visibility	0.318	0.318	0.285±0.104
Majority	0.259	0.202	0.159±0.022

Average pose nearly matches S-JEPA, while visibility reaches a pooled source-weighted macro-F1 of 0.318. These results leave open how much prediction depends on movement, average configuration, or recording conditions. The comparisons do not identify which cues S-JEPA uses. We have not shown that it discovers a particular asymmetry or needs temporal information to reach its score. The duplicated ankle feature also limits comparison with the intended angle-feature baseline; a corrected run is still needed.

Figure 4 shows where the predictions succeed and fail. S-JEPA's source-weighted recalls are 0.396 for Normal, 0.429 for MS, and 0.611 for PD; precisions are 0.568, 0.479, and 0.403. PD has the highest source-weighted recall, but its low precision shows that many PD predictions come from other labels. In raw clip counts, it gets 10/24 Normal, 8/29 MS, and 19/35 PD clips right, while assigning 13 MS clips to PD. These raw counts use different weights from the source-weighted matrix.

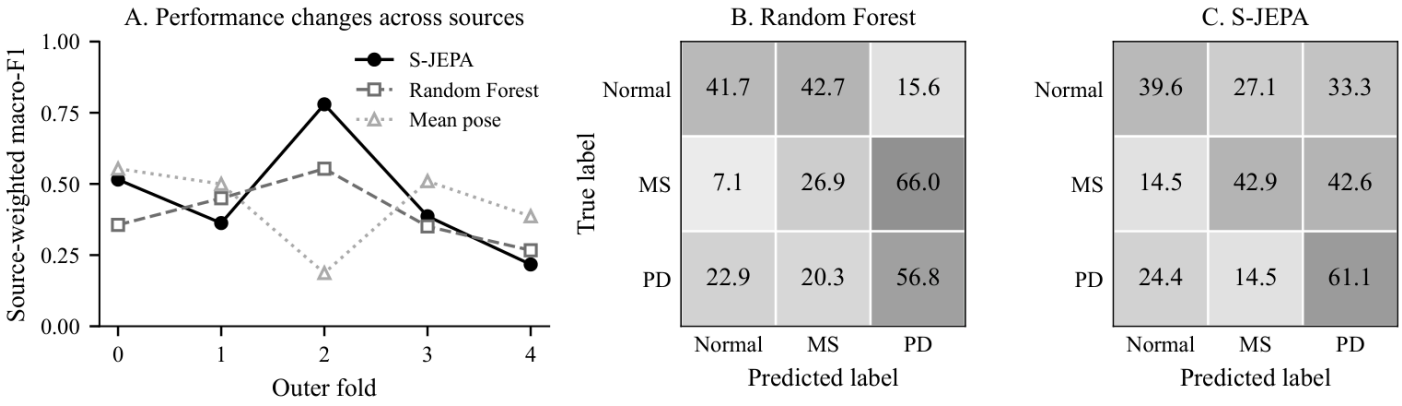


Figure 4. Held-out results. (A) Each point is an outer fold with the same test sources for all methods. Lines connect fold indices; they do not show a time trend. (B,C) Source-weighted confusion matrices, normalized within rows to percentages. Rows are supplied labels and columns are predictions. Every source contributes total weight one before normalization. Strong S-JEPA performance in fold 2 coexists with weaker folds and frequent Normal- and MS-labeled clips assigned to PD.

6.3 Evaluation Variability

S-JEPA fold macro-F1 ranges from 0.217 to 0.780, with mean 0.452 and population SD 0.189. It exceeds RF in three folds and mean pose in only one. S-JEPA performs best in fold 2: 0.780, compared with 0.553 for RF and 0.187 for mean pose. In fold 4, S-JEPA scores 0.217 and misses every 'Normal' -labeled test clip. The pooled advantage is therefore uneven across source groups. For this paper, a paired source bootstrap to the saved predictions was added. It was not in the original notebook outputs. We resampled the 16 normal, 13 MS, and 12 PD sources with replacement within label groups, using 10000 replicates and seed 20261004. Every source draw contributes a total weight of one across its clips, including repeated draws, and all methods use the same draw. The 2.5th and 97.5th percentiles give a conditional 95% interval of [0.343,0.589] for S-JEPA. The paired difference is 0.057[−0.121, 0.231] against RF and 0.016[−0.121, 0.231] against mean pose, and intervals include zero, leaving the advantage uncertain.

These intervals hold the trained models and predictions fixed. They describe source resampling variation within this collection, but exclude retraining, model selection, seed, split, and development uncertainty. Resampling treats sources as independent units, which unverified participant overlap could violate. The five folds also share training data, so they are not five independent experiments. Their SD does not provide a standard error for a simple independent-fold significance test. We leave the working null unresolved rather than claim superiority or equivalence.

7. Discussion

7.1 Interpretation of Classification Performance

The main finding is that S-JEPA did not demonstrate a consistent classification advantage over simpler pose-based summaries on this dataset. Its pooled source-weighted macro-F1 was 0.468, compared with 0.452 for mean pose and 0.411 for the Random Forest (RF) baseline. However, the difference between S-JEPA and mean pose was small, mean pose performed slightly better under equal clip weighting, and S-JEPA outperformed mean pose in only one of the five outer folds. The paired source-bootstrap intervals included zero for both comparisons, so the observed pooled differences do not establish a reliable advantage for S-JEPA.

The source-weighted confusion matrices (Figures 4B and 4C) show that S-JEPA correctly classifies MS-labeled clips 42.9% of the time, while 42.6% are classified as PD. It also classified 33.3% of Normal-labeled clips as PD. These errors may reflect overlapping gait characteristics, differences in recording conditions, or variability in the small dataset. Because the diagnostic labels were not independently verified, the results cannot establish which factors explain these misclassifications.

These results leave open whether the learned representation captures useful temporal information beyond average pose. The similar performance of S-JEPA and mean pose suggests that a more complex temporal representation was not clearly necessary to obtain the observed classification performance in this dataset. However, this comparison alone cannot determine which cues S-JEPA uses or establish that temporal information is unhelpful. The appropriate interpretation is that this experiment did not establish a consistent advantage for predictive skeleton learning, rather than that such learning cannot improve gait classification.

7.2 Optimization

Although pre-training is desirable from an optimization standpoint, its value as a representation of real-world behavior has not yet been demonstrated. The developed encoder was compared with simpler baselines, including mean pose, which does not use temporal sequences. Additional pre-training improved validation performance in only two trials. These results can be considered alongside evidence from major JEPA studies [35,36].

The training diagnostics also show that lower training loss did not consistently correspond to better validation performance. Continuation training lowered the final sampled training minibatch loss in every fold, but validation performance improved only in folds 3 and 4. The initial stage was selected in folds 0–2. This inconsistency means that additional optimization steps did not reliably improve downstream classification.

Effective-rank measurements showed variation across feature dimensions, but effective rank describes feature diversity within the measured batches, not whether the representation contains information that distinguishes Normal, MS, and PD. Similarly, the training-only silhouette scores were close to zero and did not establish test-set classification ability. These diagnostics therefore provide information about optimization and representation structure, but do not demonstrate that the encoder learned clinically meaningful features.

7.3 Temporal Effects

Positional changes within a scene can be dynamic for two reasons, making them difficult to distinguish. They may represent internal changes, such as continuous limb movement or body rotation, or they may reflect changes in viewpoint drift. The aforementioned signal may therefore reflect either gait dynamics or recording effects.

“Weighting” also has practical implications. If the number of clips varies across sources, identical predictions could lead to different performance estimates for the two systems. Therefore, the unit of evaluation needs to be defined explicitly in small-scale gait studies. In this study, S-JEPA’s pooled macro-F1 was higher than mean pose under source weighting, but slightly lower under clip weighting. This difference shows that the relative comparison depends in part on how recordings are weighted.

7.4 Limitations

A major limitation is label accuracy. These labels have not been clinically validated and may include multiple source identifiers for the same individual, meaning that clips may not represent independent observations. S-JEPA training did not proceed entirely as planned. Although the transformation reflected the context and swapped left and right landmark positions, the mask position remained unchanged. In a seed-42 audit of 512 masks, 17,931 (21.0%) of the 85,409 target slots would have their coordinates exposed if the window were reflected. Although this information remains within the training sample and does not cross the train-test source boundary, its impact on the training process has not been measured.

The heterogeneous recording conditions also limit interpretation because viewpoint, resolution, frame rate, occlusion, and pose-estimation errors may affect the extracted skeletons. In addition, the preprocessing centers each frame on the hip midpoint and scales it by shoulder-to-hip distance. This removes absolute pelvis motion and apparent torso size, and frame-to-frame scale changes can alter measured displacement. Consequently, the input does not preserve every physical gait quantity exactly, and model predictions should not be interpreted as direct measurements of walking speed or stride length.

Finally, the paired source-bootstrap intervals hold the trained models and predictions fixed. They do not account for retraining, model-selection, seed, or split variability. The five outer folds also share

training data, so they should not be treated as five independent experiments. These factors limit the strength and generalizability of the performance comparisons.

7.5 Future Work

Statistics and individual performance metrics serve as descriptive comparisons. Furthermore, trained models should be compared against untrained ones using the same architecture to investigate whether temporal training contributes beyond the architecture itself. A controlled comparison with temporally shuffled or otherwise order-invariant inputs could further test whether temporal order contributes information beyond individual poses. Source-based results should also be corroborated by findings that account for temporal order and source grouping. The Random Forest feature-extraction defect should be corrected before its performance is used as a definitive comparison with S-JEPA.

To determine whether the clinically motivated landmark selection itself contributes to performance, future experiments should compare this mask with alternative anatomical and motion-magnitude masks under otherwise identical training and source-grouped evaluation conditions. The reflection-mask mismatch should also be corrected or explicitly controlled. Clinical studies require verified labels and independent datasets. Evaluation on an independent dataset with verified diagnostic labels and participant identities, using participant-level separation where possible, would provide a stronger test of whether the method generalizes beyond the current video sources.

8. Conclusion

This study evaluated a pipeline for applying Skeleton Joint Embedding Predictive Architecture (S-JEPA) to neurological gait classification using video-derived skeleton sequences. Using 88 gait clips from 41 distinct YouTube sources, we compared S-JEPA with simpler pose-based representations and a Random Forest baseline. When each source was given equal total weight, S-JEPA achieved an average F1 score of 0.468, compared with 0.452 for mean pose and 0.411 for Random Forest. However, mean pose performed better when all clips were weighted equally and outperformed S-JEPA in four of the five evaluation folds. Bootstrap analysis did not establish a reliable advantage for S-JEPA over either baseline.

These findings demonstrate the feasibility of evaluating predictive skeleton representations for neurological gait classification in a small, source-grouped video dataset, but do not establish a consistent advantage over simpler representations. The results should also be interpreted cautiously because the diagnostic labels were not independently verified and the dataset was limited in size. Future work should evaluate the pipeline on larger datasets with verified diagnostic labels and participant identities, and investigate whether alternative training strategies improve classification performance.

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